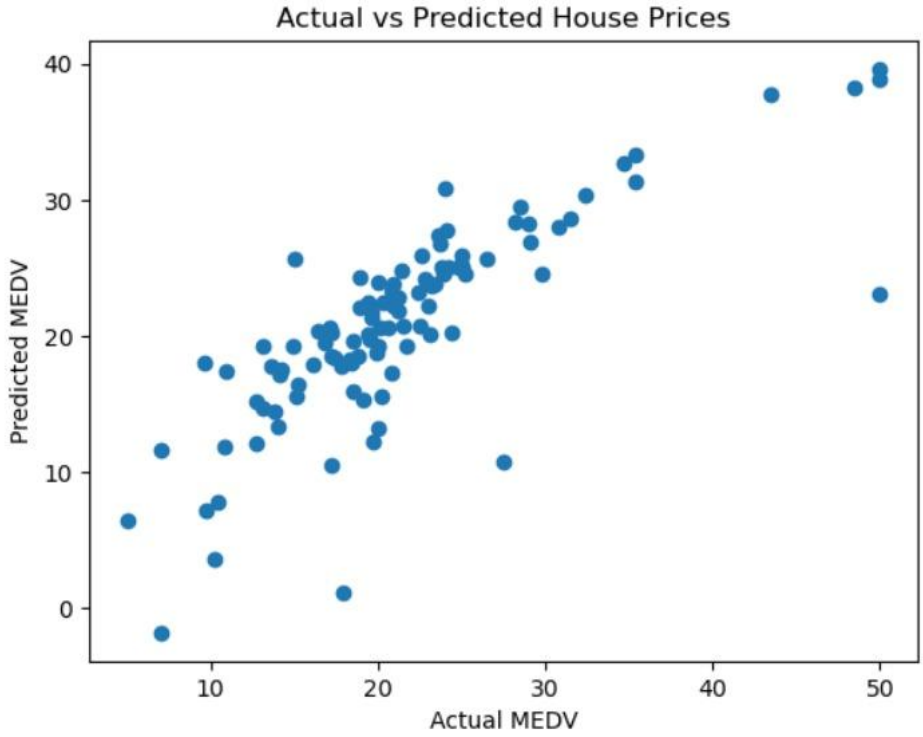
TASK 2

A MULTIPLE LINEAR REGRESSION MODEL

A real estate analyst wants to predict the median value of houses in different neighborhoods based on various property and neighborhood features. You are tasked with building a Multiple Linear Regression model that can predict housing prices using numerical predictors

CODE AND OUTPUT :

```
import numpy as np
import pandas as pd
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split

df = pd.read_csv("Boston.csv")

df = df.dropna()

X = df[['RM', 'LSTAT', 'PTRATIO', 'INDUS', 'NOX', 'AGE']]
y = df['MEDV']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

model = LinearRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print(f"Coefficients: {model.coef_}")
print(f"Intercept: {model.intercept_}")

from sklearn.metrics import mean_squared_error, r2_score

mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)
```

```python
print(f"Mean Squared Error: {mse}")
print(f"R^2 Score: {r2}")

import matplotlib.pyplot as plt

plt.scatter(y_test, y_pred)
plt.xlabel("Actual MEDV")
plt.ylabel("Predicted MEDV")
plt.title("Actual vs Predicted House Prices")
plt.show()
```

```
Coefficients: [ 4.7768173  -0.59349623 -0.94868975  0.06578322 -8.65012257  0.03227307]
Intercept: 19.276964533413906
Mean Squared Error: 26.782523432699985
R^2 Score: 0.6347860115964862
```